AIVOA.AI

Software Testing & QA Challenge

Product Thinking • Exploratory Testing • User-Centric QA

Application	QMS.AIVOA.AI – AI-Native Quality Management System
Module Tested	Quality Events → Deviation → New Record
Environment	http://216.48.184.249:5274/
Test Account	testing@aivoa.ai
Site Context	SIT-001 – MOXIFLOXACIN PLANT 1 | API (primary)
Report Date	October 03, 2026
Role	Quality Assurance Engineer (QAE)

This report demonstrates deep product understanding of pharmaceutical QMS workflows, rigorous exploratory testing of the Deviation module, documented defects with user-impact analysis, and strategic 10X product recommendations.

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2. Executive Summary

This submission addresses the AIVOA.AI Software Testing & QA Challenge with a dual focus: (1) demonstrating genuine product thinking about how a pharmaceutical Quality Management System is used during Active Pharmaceutical Ingredient (API) manufacturing, and (2) performing user-centric exploratory testing of the Deviation workflow in the live application.

Key Outcomes:

- Mapped a realistic API manufacturing scenario (Moxifloxacin API) linking raw material receipt through deviation, investigation, CAPA, batch release, and post-market signals.
- Explored the full Deviation Initiation form and list view under the site context SIT-001 – MOXIFLOXACIN PLANT 1.
- Identified multiple functional, validation, UI/UX, and security-related issues with clear user-impact statements.
- Proposed six high-leverage product improvements that combine workflow intelligence, AI assistance, traceability, and compliance automation.

The evaluation prioritizes **how a Quality user thinks and works** over checklist testing of individual fields.

3. QMS Understanding – API Manufacturing Context

3.1 What is a QMS and Why It Is Needed

A Quality Management System (QMS) in the pharmaceutical industry is the formalized system that documents processes, procedures, and responsibilities for achieving quality policies and objectives. For API manufacturing it is not optional — it is the backbone that demonstrates control over the process and product to regulators (FDA, EMA, MHRA, WHO, etc.) under frameworks such as ICH Q7 (Good Manufacturing Practice for Active Pharmaceutical Ingredients), 21 CFR Part 211, and Schedule M.

Why it is needed: An API is the pharmacologically active substance. Any uncontrolled variation in raw materials, process parameters, equipment, environment, or analytical methods can compromise the safety, identity, strength, purity, and quality (SISPQ) of every downstream finished dosage form. The QMS provides the structured evidence trail that deviations were detected, investigated, corrected, and prevented from recurring, and that batches were released only after justified disposition.

3.2 End-to-End Process Map: Raw Material → Batch Release

Realistic scenario used: Manufacture of Moxifloxacin Hydrochloride API at Plant SIT-001.

Stage	What Happens	Who Uses It	QMS Touchpoint
Raw Material Receipt & Quarantine	Incoming solvents, intermediates, catalysts sampled, tested against specs, released or rejected.	Warehouse, QC, QA	Supplier Quality, CoA Management, Sampling
Manufacturing (Reaction, Isolation, Purification)	Multi-step synthesis under controlled parameters (temp, pH, time, agitation).	Production, Engineering	Batch Records, In-Process Checks, Equipment Logs
In-Process Checks (IPC)	Critical quality attributes monitored at defined stages.	Production, QC	OOS / OOT triggers, Deviation if limit exceeded
Deviation	Any departure from approved process, procedure, or specification.	Any employee → QA ownership	Deviation module (core focus)
Investigation & Root Cause	Structured analysis (5-Why, Fishbone, FMEA) to find true cause.	QA, Production, SME, Cross-functional team	RCA tools, Linked records

CAPA	Corrective action removes cause; Preventive action stops recurrence.	QA, Department owners	CAPA Manager, Effectiveness Checks
Batch Release / Disposition	QA decides: Release, Reject, Reprocess, or further investigation.	QA Head / Qualified Person	Batch disposition linked to closed deviations/CAPAs
Complaint / Recall / AE	Post-market signals may loop back to manufacturing root causes.	Pharmacovigilance, QA, Regulatory	Complaints, Product Recalls, Adverse Events modules
Supplier Quality	Vendor audits, material quality issues, change notifications.	Procurement, QA Supplier Management	Supplier Quality records, linked deviations

Table 1: End-to-end QMS process map for API manufacturing (Moxifloxacin example)

3.3 Focus: Deviation → Investigation → Root Cause → CAPA → Closure

Deviation example (realistic): During the crystallization step of Moxifloxacin API Batch MET-2026-042, reactor temperature exceeded the approved upper limit of 25 °C for 47 minutes (observed peak 31.2 °C). Immediate containment: process paused, material quarantined, temperature excursion logged, QA notified.

Investigation: Cross-functional team reviews batch record, equipment logs, SCADA trends, operator interviews, maintenance history, and previous similar events. Tools applied: 5-Why, Fishbone (man, machine, method, material, measurement, environment).

Root Cause: Cooling water supply valve actuator intermittent failure (latent equipment defect) combined with inadequate real-time alarm acknowledgment procedure.

CAPA: Corrective – replace actuator, recalibrate temperature loop, reprocess or reject material based on impact assessment. Preventive – update preventive maintenance schedule, add secondary temperature sensor with independent alarm, revise SOP for excursion response, train operators.

Closure: Effectiveness check after next three batches shows no recurrence; CAPA closed; deviation closed; batch disposition finalized; change control opened for SOP/equipment updates if needed.

3.4 Connections: Complaints, Recalls, Adverse Events & Supplier Management

A well-designed QMS treats these as interconnected signals rather than isolated modules:

- A customer complaint about impurity profile can open a retrospective deviation investigation on historical batches.
- An adverse event report may trigger a health-hazard evaluation that leads to a recall decision, which itself generates CAPAs.
- Supplier quality failures (e.g., off-spec solvent) frequently appear first as deviations or OOS results.
- AIVOA already surfaces modules for Complaints, Product Recalls, Adverse Events, and CAPA Manager — the power lies in bidirectional linking and AI-assisted correlation.

3.5 What AIVOA's Deviation Workflow Is Trying to Achieve

From the observed stages (Initiation → QA Review → RCA & CAPA → Closure) and the site context of an API plant, AIVOA's Deviation workflow aims to:

1. Capture the event rapidly with structured mandatory data (title, initiator, date, category, risk, impacted items).
2. Enforce a gated lifecycle so that investigation and CAPA cannot be skipped.
3. Maintain an audit trail (21 CFR Part 11 claims visible on login) with electronic signatures and activity logging.

4. Provide a single source of truth for quality events that can later feed APR/PQR, management review, and regulatory inspections.

5. Leverage AI (implied by the product name and positioning) to accelerate investigation, suggest root causes, and recommend CAPAs — though this intelligence layer was not fully exercised in the Initiation form under test.

4. Testing Approach & Methodology

Mindset adopted: “If I were a Quality user in a pharmaceutical company, what would I expect this workflow to do?”

Testing was exploratory and user-journey oriented rather than pure field-by-field checklist. Coverage included: happy-path creation, required-field behavior on Save Draft, navigation between list and form, presence of existing records (including suspicious XSS titles), workflow stage visibility, UI consistency, and edge considerations around incomplete data, repeated actions, and browser refresh.

Test Environment Observations

- Dashboard shows 298 Active Deviations, 42 Open CAPAs — high volume environment typical of a multi-product API site.
- Deviation List supports filtering by Status (Draft, Under Review, Closed, Cancelled) and Workflow Stage (Initiation, QA Review, RCA & CAPA, Closure).
- New Record opens a multi-stage form with locked future stages (padlock icons on QA Review, RCA & CAPA, Closure).
- Rich-text editor present for Immediate/Containment Action and Prescribed Parameter/Specification.
- Impacted Items grid allows multiple rows with Item Category, Item ID/Reference, Impact Description.

5. Exploratory Test Results – Deviation New Record

5.1 Observed Form Structure (Initiation Stage)

Field	Required?	Notes
Title	Yes (*)	Auto-fills 'Untitled Record' on empty Save Draft
Initiated By	Yes (*)	User selector
Date of Event	Yes (*)	Date-time picker (mm/dd/yyyy)
Shift	No	Dropdown
Deviation Category	Yes (*)	Dropdown
Department Owner	Yes (*)	Department selector
Stage of Event	No	Dropdown
Initial Risk Assessment	Yes (*)	Dropdown
Immediate / Containment Action Taken	No	Rich-text editor
Impacted Items & References	Yes (*)	Grid – at least one row expected
Prescribed Parameter/Specification	Visible	Rich-text (further down form)

5.2 Key Observations from Exploration

- Save Draft succeeds with zero required fields completed; system assigns DEV-xxx number and default title “Untitled Record”.
- Workflow stages are visually gated (padlocks), which is good for process control.
- Existing records include titles that appear to be XSS test payloads (e.g., <script>alert('xss')</script>), indicating prior security testing or residual data.
- List view is clean, filterable, and shows status/stage badges clearly.
- Activity / chat-style panel at bottom of form suggests collaboration features.

6. Bug Report (Detailed)

Each bug is documented from the perspective of a pharmaceutical Quality user. Severity is relative to compliance risk, data integrity, and operational efficiency.

BUG-01: Required-Field Validation Missing on Save Draft

Steps to Reproduce:

1. Navigate to Quality Records → Deviation List → + New Record.
2. Leave all fields empty (Title, Initiated By, Date of Event, Category, Department Owner, Risk, Impacted Items).
3. Click “Save Draft”.

Expected: System should either (a) prevent save and highlight mandatory fields, or (b) allow draft but clearly flag incomplete status and block progression to next stage until completed.

Actual: Draft is created successfully as DEV-574 (or next number) with title “Untitled Record”. No validation messages appear.

Why it is a bug from the user’s perspective: In a regulated environment, an incomplete deviation record that looks “saved” creates false confidence. A Quality reviewer opening the list sees many “Untitled Record” drafts and cannot distinguish intentional work-in-progress from accidental empty saves. During an inspection, a large number of empty drafts can raise questions about data integrity culture and system design. Risk: audit findings, delayed investigations, orphaned records that never progress.

BUG-02: Potential Stored XSS in Deviation Title (Observed in Existing Data)

Observation: Deviation List contains record DEV-570 with title rendered as: <script>alert('xss')</script>.

Expected: All user-controlled text (especially titles displayed in lists and headers) must be properly sanitized/escaped so that script tags are treated as plain text.

Actual: The payload appears stored and visible in the list. Whether it executes depends on client-side rendering; the mere presence indicates insufficient input sanitization or encoding at rest/display.

Why it is a bug from the user’s / security perspective: In a 21 CFR Part 11 system that claims audit-trail protection, any ability to inject executable content is a serious integrity and security risk. An attacker or even a careless user could craft titles that disrupt the UI, steal session data, or escalate privileges. For Quality users, it erodes trust that the system protects the electronic records they rely on for batch release decisions.

BUG-03: No Immediate Visual Feedback / Blocking of Progression for Incomplete Critical Fields

Steps: After saving an empty draft, attempt to understand how to move to QA Review stage.

Expected: Clear indicators (progress meter, checklist, disabled “Submit for QA Review” button with tooltip explaining missing mandatory data).

Actual: Stages remain locked with padlock icons; no guidance on what is still required to unlock the next stage. User must discover by trial.

User impact: New or infrequent users of the system waste time, increase support tickets, and may leave records in permanent Draft state. In high-volume plants (298 active deviations observed), this friction multiplies into delayed investigations and potential missed regulatory timelines.

BUG-04: Date of Event Control Accepts Ambiguous / Future / Invalid Formats Without Clear Constraint

Observation: Date picker shows format mm/dd/yyyy with time component. No visible max-date = today constraint or clear error handling for future dates or nonsense values during exploratory interaction.

Expected: Date of Event cannot be in the future; past dates beyond a configurable retention window may need warning; clear validation message on blur/submit.

User / compliance risk: An event dated in the future or with incorrect year can corrupt trending, APR/PQR data, and investigation timelines. Inspectors frequently examine whether systems enforce logical data constraints.

BUG-05: Impacted Items Grid – Required Indicator vs. Empty Row Behavior

Observation: Section is marked required (*), yet an empty row is present by default. Saving draft with empty Item Category / Reference / Description succeeds.

Expected: Either the grid requires at least one fully completed row before any save that advances state, or the UI clearly distinguishes “optional until submit” from “mandatory for draft completeness”.

User impact: Quality users may believe they have documented impact when in reality the linkage to batches, equipment, or materials is missing. Traceability from deviation to affected batches is a core inspection focus; missing impact data weakens the entire record.

BUG-06: UI Density / Scroll Experience on Long Form

The Initiation form is long (multiple sections + rich-text + grid). Scrolling behavior and sticky headers / progress indicators are limited. On smaller viewports or when collaboration chat is open, critical action buttons (Save) can be less discoverable.

User impact: Operators or QA specialists working under time pressure (e.g., during a real excursion) need fast, low-friction capture. Friction increases the chance of incomplete or delayed reporting.

7. 10X Product Thinking – Recommended Improvements

These recommendations are designed to move AIVOA from a capable electronic QMS to an intelligent, user-delighting, inspection-ready platform that reduces cognitive load and accelerates quality decisions.

1. AI-Assisted Deviation Intake & Smart Defaults

When a user starts a new Deviation, allow natural-language description ("Reactor temperature went to 31 °C for 47 min on Batch MET-2026-042 during crystallization"). AI parses it, pre-fills Title, Category, Stage of Event, Impacted Items (batch + equipment), suggests Initial Risk, and drafts Immediate Action. User confirms/edits. This turns a 10–15 minute form into a 60-second review — especially valuable for production staff who are not full-time QA users.

2. Contextual Investigation Workspace with Linked Evidence Graph

In the RCA & CAPA stage, surface a living graph of related records: same equipment, same batch family, same material lot, prior similar deviations, open CAPAs, recent change controls, and supplier issues. AI proposes candidate root causes ranked by historical pattern match. One-click attach of SCADA trends, batch record pages, or CoAs. This dramatically improves investigation quality and reduces "tribal knowledge" dependency.

3. Real-Time Completeness & Progression Guardrails

Replace silent padlocks with an explicit completeness score and checklist visible at all times. "Submit for QA Review" becomes enabled only when mandatory fields + at least one impacted item + containment action are present. Hovering the disabled button lists exactly what is missing. Optional "Save Incomplete Draft" remains available but is visually distinct and cannot be advanced.

4. Automated Effectiveness Check Scheduling & AI Effectiveness Prediction

When a CAPA is closed, the system auto-creates Effectiveness Check tasks at configurable intervals (e.g., after next 3 batches or 90 days). AI monitors subsequent deviations/OOS for the same root-cause signature and proactively alerts if the CAPA appears ineffective — turning CAPA from a paperwork exercise into a living control.

5. Cross-Module Signal Correlation Dashboard (Complaints ↔ Deviations ↔ Recalls ↔ Supplier)

A dedicated "Quality Signal Network" view that automatically surfaces possible links between a new complaint, historical deviations on the same product/process, supplier quality events, and open CAPAs. One-click initiation of a linked investigation or health-hazard evaluation. This is the difference between reactive silos and proactive quality intelligence — and a strong differentiator for AI-native positioning.

6. Inspection-Ready Export Pack & Narrative Generator

One-click generation of a complete deviation investigation package (PDF + audit trail + linked CAPAs + effectiveness evidence) formatted for FDA/EMA inspection style, including an AI-drafted executive summary that a QA reviewer can edit. Reduces the frantic pre-inspection scramble that every quality team knows too well.

8. Risk Assessment & Compliance Notes

The application positions itself with 21 CFR Part 11, Schedule M, and SOC 2 claims. From a testing perspective:

- Audit trail and electronic signature capabilities are asserted; detailed verification of signature meaning, non-repudiation, and trail completeness was outside the scope of the Initiation form test but should be a priority for formal validation.
- Input sanitization (XSS) requires hardening before production use in a regulated environment.
- Draft behavior must be carefully defined in the validation master plan so that incomplete records cannot be mistaken for controlled records.
- High volume of existing deviations (298 active) underscores the need for excellent filtering, bulk actions, and AI prioritization to keep the system usable.

9. Conclusion & Recommendations

AIVOA.AI demonstrates a modern, multi-module QMS with a clear Deviation lifecycle that mirrors real pharmaceutical practice. The UI is contemporary and the stage-gated design is directionally correct for compliance.

The highest-value next steps are: (1) enforce meaningful completeness rules while preserving flexible draft capability, (2) eliminate any residual XSS vectors, (3) invest in the AI-assisted intake and investigation workspace that the product name and positioning promise, and (4) close the loop between deviations, CAPAs, complaints, recalls, and supplier quality with intelligent correlation.

From a senior QAE standpoint, the product has strong foundations. The recommended 10X features would transform it from a solid electronic system into a genuine quality intelligence platform that Quality professionals would actively prefer to use — not merely be required to use.

10. Appendix – Test Artifacts & Notes

- Application URL: http://216.48.184.249:5274/
- Login: testing@aivoa.ai / password123
- Site context during test: SIT-001 – MOXIFLOXACIN PLANT 1 | API (primary)
- Created draft example: DEV-574 – Untitled Record (empty required fields)
- Observed existing XSS-titled record: DEV-570
- Dashboard metrics observed: 42 Open CAPAs, 298 Active Deviations, 2 Open Audits, 0% CAPA Effectiveness (displayed)

This document constitutes the written deliverable covering QMS Understanding, Deviation Workflow analysis, Bug Report, and 5+ Product Improvements. Video demonstrations (Testing Demo + QMS Understanding) are intended to accompany this report in the formal submission package.